Semester: 5th
Subject Name:- DMDW & Code:-CS30013
Branch (s): - CSE

AUTUMN MID SEMESTER EXAMINATION-2024

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University

DATA MINING AND DATA WAREHOUSING
[CS30013]

Time: 1 1/2 Hours **Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

a) What is data transformation? Explain the data transformation of 20 using z-score normalization for the data points {10,15, 20, 50, 60}.

b) Suppose a group of 12 sales records have been sorted as follows:
5,10,11,12,15,35,50,55,72,92,204,215
partition them into three bins using equi-depth partitioning then smooth them using bin means.

c) Briefly discuss the terms Data Mart, Data cube and Data Warehouse.

d) In the Apriori algorithm, what happens if the minimum support threshold is set too low or too high?

e) What are the non-trivial costs of using Apriori algorithm.?

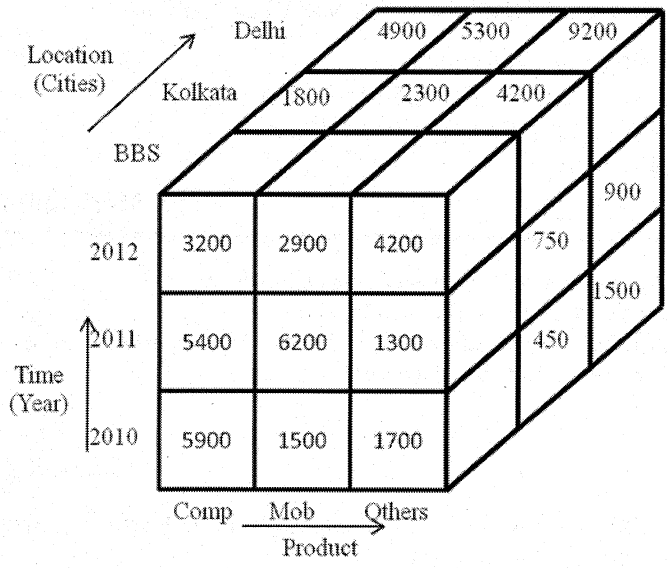
2. What is Bessel's correction? Calculate and draw a Box plot for a dataset {35, 42, 38, 45, 48, 60, 55, 40, 50, 53, 41, 36}. Determine if there are any potential outliers in the dataset using the IQR method. If there are outliers, identify them and provide their values. [5 Marks]

3. a. Discuss briefly the different types of problem occurs during data integration? [2 Marks]
b. Calculate the correlation coefficient between the following values of demand and the corresponding price of commodity? Describe the significance of this result. [3 Marks]

Demand	65	66	67	67	68	69	70	72
Price	67	68	65	68	72	72	69	71

4. a. Briefly describe how the components of data warehouse architecture interact to support decision-making with diagram. [3 Marks]

b. Solve the following queries by taking this cube for SALES using OLAP operations. Consider your own concept hierarchy for different dimensions. [2 Marks]



i. Find the total number of Mob and Comp sold in 2012 in different regions (North or East).
ii. Find the different products sold in 2011 in different cities.

5. Consider the transaction data set of a Mall and perform the following: [5 Marks]

i. Find all the frequent 1-itemsets, 2-itemsets, and 3-itemsets using a minimum support threshold of 30%. Use the Apriori algorithm to list all candidate itemsets and identify the frequent ones.
ii. Based on the frequent itemsets identified, generate association rules with a confidence level of at least 50%. For each rule, calculate support, confidence, and lift.

TID	**Items Purchased**
1	Apples, Bananas, Grapes, Milk
2	Bananas, Grapes, Bread, Butter
3	Apples, Bananas, Bread
4	Milk, Bread, Butter
5	Apples, Bananas, Milk, Butter
6	Bananas, Grapes, Butter
7	Apples, Milk, Bread
8	Grapes, Milk, Butter, Bread
9	Bananas, Milk, Butter, Grapes
10	Apples, Bananas, Grapes

*** Best of Luck ***